A Price-Time Priority Order Book Simulator, a Controlled Liquidity Stress Experiment, and the Cost of Ignoring the Queue

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https://github.com/mateeniqbal3/lob-simulator

Abstract

I build a limit order book matching engine with exchange-style price-time priority, in-place partial fills that keep queue position, and per-order lifecycle accounting. I verify it with targeted tests, randomized invariant checks after every event, a differential test against an independent brute-force matcher, and a bug-reintroduction check in which all 15 injected bugs were caught. I then use it for two experiments on a calibrated zero-intelligence order-flow model, both designed in advance.

In the first, a 5-minute burst of market sells at five times the baseline rate, compared with same-seed no-event controls over 20 seeds, widens the spread by a median 1.18 ticks and removes about half of top-of-book bid depth, in all 20 seeds. Liquidity then recovers in 27–54 s. I argue that this speed is a mechanical consequence of the model's exogenous liquidity supply, not a finding about markets.

In the second, passive probe orders run on identical background flow show that the textbook naive assumption, instant fill at the posted price, reports 1.40 ticks of spread earned per order. The queue-aware engine shows the same orders paying 0.26 ticks net. About half of the gap comes from queue position and half from the price never reaching the order. All results come from synthetic order flow and are a methodology demonstration, not evidence about any real market.

Keywords: limit order book, matching engine, price-time priority, market microstructure, liquidity resilience, queue position, fill probability, backtesting, simulation

JEL classification: C63, D47, G14

Framing. Every result in this writeup comes from synthetic order flow generated by a documented zero-intelligence model, with a deliberately designed stress scenario. It is a methodology demonstration. Nothing here describes, or should be read as evidence about, any real market, instrument or date.

1. Motivation

Two questions come up whenever a simulated or backtested execution result is presented:

- **Is the matching mechanism right?** Many plausible-looking simulations quietly fill orders out of priority, move partially filled orders to the back of the queue, or create and destroy volume. The rest of the analysis inherits any such error.
- **Would the orders actually have filled?** Backtests of passive strategies routinely assume that a posted order fills when, or even before, the price touches it. That ignores the queue already waiting at that price.

This project treats the first question as the primary engineering deliverable and the second as the primary empirical one. The event study sits between them. It checks that the simulated book responds to stress in the direction microstructure reasoning predicts, and it measures how fast.

Related work. Gould et al. (2013) survey limit order books and the models used to simulate them, and Bouchaud et al. (2018) give a book-length treatment of order flow, impact and resilience. Zero-intelligence models (Smith et al., 2003; Cont, Stoikov & Talreja, 2010) show that many statistical features of the book follow from the matching mechanism alone; the queue-reactive model of Huang, Lehalle & Rosenbaum (2015) makes order arrival depend on queue sizes. Resilience after a large trade is central to optimal-execution models such as Obizhaeva & Wang (2013), and the cost of being filled when better-informed flow is trading against you goes back to Glosten & Milgrom (1985). This project contributes no new model. It contributes a verified engine and a controlled measurement of how much a common backtesting shortcut overstates passive execution quality.

2. Matching engine

State. Each side of the book is a sorted list of occupied prices (integer ticks). Each price holds an insertion-ordered dictionary of resting orders. Insertion order *is* time priority. Cancelling any order is O(1), and a partial fill or a size reduction changes an order's quantity in place, so its position cannot change.

Rules.

- An incoming order trades against the best opposite price first, and within a price against the earliest order.
- The trade price is the resting order's price.
- A limit order's remainder rests; a market order's remainder expires.
- Cancels of unknown, filled or already-cancelled orders are rejected and counted rather than raised, because in live flow a cancel can race a fill.
- Invalid input (a non-positive quantity, a non-integer price, a duplicate id, an out-of-order timestamp) raises.

The book itself refuses a crossing insert, so a matching bug that failed to match a marketable order would fail loudly, not persist as a crossed book.

Verification (`docs/engine_correctness.md`):

1. The five properties required by the specification, each with a hand-built book: price priority,

time priority, partial-fill queue preservation, cancellation, and no look-ahead.
2. Sixty seeded adversarial random streams. They use a narrow price band, so many orders are marketable, plus exact timestamp ties, oversized market orders, and partial, full and unknown cancels. Per-order accounting (`qty = filled + cancelled + expired + resting`), side-level volume conservation and full structural invariants are checked after every event.
3. A differential test. On the same streams, every fill and the final book must match a deliberately naive matcher that re-sorts the book before every match and shares no code with the engine.
4. A bug-reintroduction harness, which injects 15 specific bugs one at a time: LIFO queues, a partial fill sent to the back, double credit on fills, a missing time-order guard, and others. Each one fails the test written for it.

Known gaps are listed rather than implied away:

- two properties are guarded by a single test each;
- 15 mutations are a sample;
- the engine has never seen real exchange messages.

3. Order-flow model and calibration

The model is zero-intelligence, in the spirit of Smith et al. (2003) and Cont, Stoikov & Talreja (2010):

- limit orders arrive per side at 1.0/s;
- a buy is placed `d` ticks below the best ask (a sell `d` ticks above the best bid), with $P(d) \propto d^{-0.5}$ on 1–10 ticks, so limit orders never cross;
- market orders arrive per side at 0.20/s;
- each resting order is cancelled at 0.05/s;
- limit sizes are 100 × U{1..5} shares and market sizes 100 × Geometric(0.4).

The simulation is exact (competing Poisson clocks). It restarts the clock at every change in rates, so nothing before a scheduled event depends on it.

Calibration used no-event runs on seeds reserved for the purpose. The first three parameter sets produced a book that was too deep and too static: the spread was at 1 tick 98–100% of the time, and the mid moved 2–10 ticks in 90 minutes. I then wrote down four targets:

1. a mean spread of 1.2–3 ticks;
2. mean best-bid depth of 2–5× the mean market order;
3. both sides non-empty more than 99.9% of the time;
4. a moving mid.

Two of four further configurations met all of them. I chose the one with rounder parameters and fewer events. It has a mean spread of 1.34 ticks, best-bid depth 2.7× the mean market order, and a 42-tick mid range. All seven configurations are listed in ADR-001.

4. Experiment 1: controlled liquidity stress

Design (fixed in ADR-002 before any stressed run):

- **Scenario:** the sell market-order rate goes from 0.20 to 1.00/s for 300 s at t = 3,600 s. Nothing else changes.
- **Windows:** 30 minutes of warm-up are discarded, then there is a 30-minute pre-event window and a 55-minute post-event window.
- **Replication:** twenty seeds, each run with and without the event. The two runs are event-for-event identical before 3,600 s, so each effect is a paired (stressed − control) difference.
- **Metrics:** spread, best-bid depth and best-5-level bid depth, sampled every second as of that second.
- **Recovery time:** from the end of the event until the trailing 30 s mean is back within 10% of the run's own pre-event mean. The same rule applied to the control run gives a placebo.

Results.

- **During the event:** the median spread was 2.45 ticks against 1.28 (+1.18, IQR 1.02–1.36, 20/20 seeds). Best-bid depth was 349 shares against 705 (20/20), and best-5 bid depth 1,474 against 3,346 (20/20).
- **The spread effect is understated:** the bid side was completely empty in a median 6% of event samples, where the spread is undefined and excluded.
- **Ask side:** it thinned as well (−272 shares at the best ask). That is plausibly because new asks are placed relative to a falling bid, though I did not verify the mechanism.
- **Price:** the mid fell a median 99.5 ticks and, with no fundamental value in the model, did not revert.
- **Recovery:** median 27 s for the spread, 54 s for best-bid depth and 38 s for best-5 depth. The placebos were 0 s, 2 s and 0 s. No run was censored. Five minutes after the event, window means were indistinguishable from control.
- **Robustness:** across the pre-registered grid (2×/5×/10× × 60/300/900 s), the direction of every during-event effect holds in every cell. The effects scale with intensity (spread +0.25 to 0.31, +1.07 to 1.19, +1.89 to 2.09 ticks) but barely with duration. Median recovery never exceeds a minute. At 2× it is hard to tell apart from placebo noise, and at 10× the bid side is empty for about a third of the event.

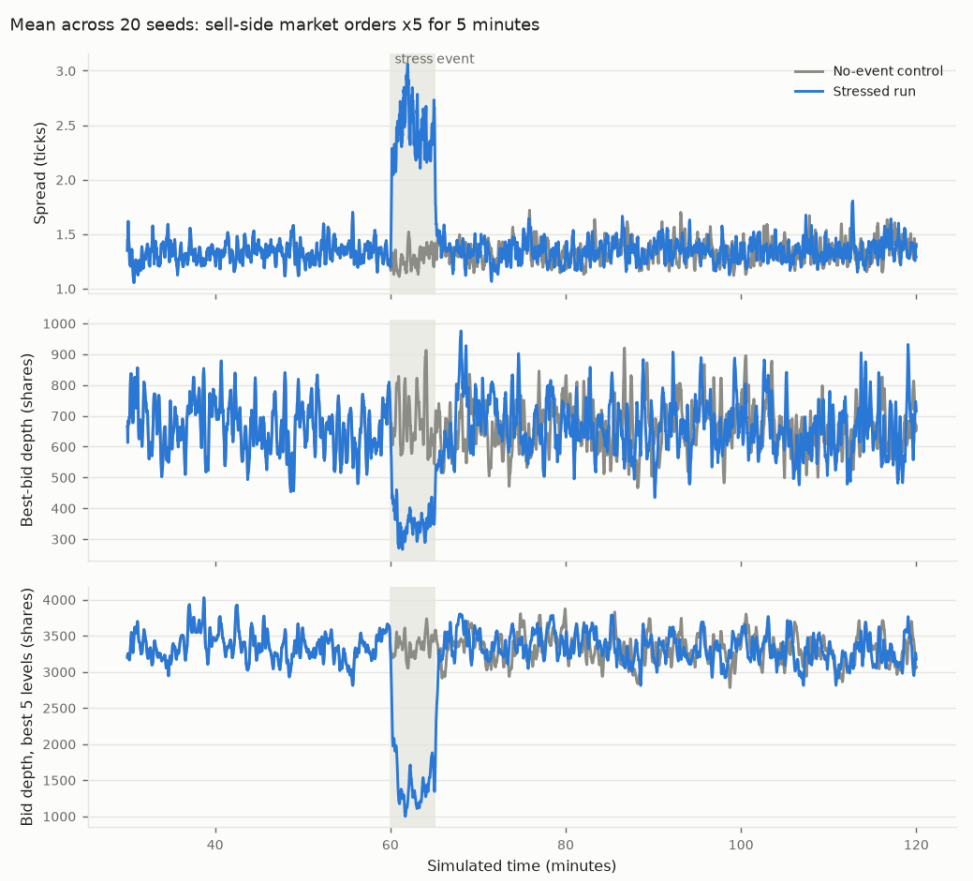


Figure 1. Spread, best-bid depth and best-5 bid depth, stressed versus same-seed control runs, averaged over 20 seeds.

Interpretation. The *direction* of each effect is what microstructure reasoning predicts. The *speed* of recovery is not what is typically argued for real markets. There, resilience depends on liquidity suppliers who may withdraw or reprice after adverse flow; see Foucault, Kadan & Kandel (2005) for a model in which resilience is endogenous. In this model, three things drive recovery mechanically:

- liquidity supply is exogenous, so limit orders keep arriving at the baseline rate;
- orders are placed relative to the opposite quote, so a wide spread is refilled within seconds;
- resting orders live for about 20 s, so the book forgets the shock.

The result therefore validates the simulator's qualitative response. It is not an estimate of how quickly real books recover. The obvious extension is to make limit-order intensity respond to recent adverse flow and measure the change in recovery time.

5. Experiment 2: naive vs. queue-aware execution

Design (fixed in ADR-008 before running):

- **Probes:** every 15 s, a 100-share buy at the best bid and a 100-share sell at the best ask, each patient for 60 s, then completed by market order. This runs on the primary stressed flow, for 14,257 probes over 20 seeds.
- **Three assumptions on the same background event log:**
  - *naive*: instant fill at the posted price;
  - *price-touch*: fill when any trade prints at or through the posted price, ignoring the queue;
  - *queue-aware*: the probe is actually inserted into the engine, behind every order already at its price.

Results.

	Naive	Price-touch	Queue-aware
Fill rate	100%	94.8%	86.9%
Price vs. posted (ticks)	0.00	+0.43	+0.84
Effective spread (ticks)	−1.40	−0.59	+0.26
Median time to fill	0 s	3.3 s	10.6 s
60 s markout (ticks)	+0.73	+0.31	−0.13

Effective spread is 2 × side × (execution price − mid at posting), with side +1 for buys and −1 for sells; the execution price averages the passive fill and any market-order completion. Negative values mean spread was earned. A positive markout means the mid later moved in the probe's favour.

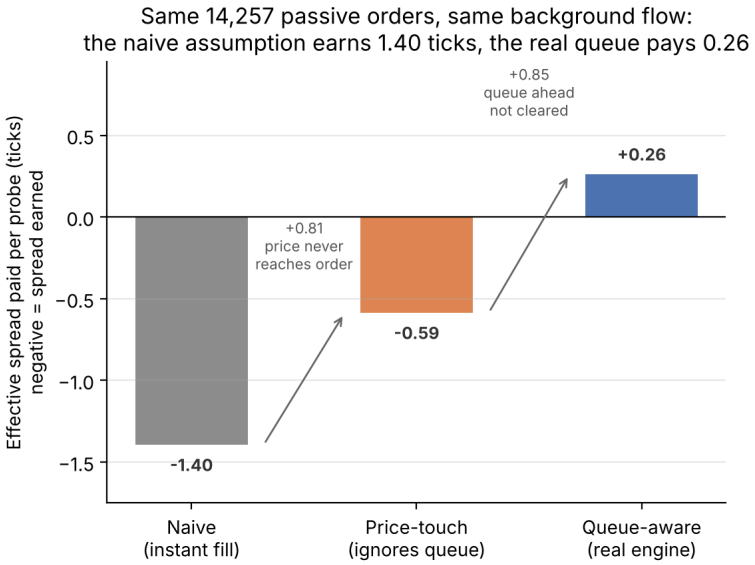


Figure 2. Effective spread per probe under the naive, price-touch and queue-aware fill assumptions.

Mechanism. The middle column separates two effects:

- **Price reach.** A posted order may simply never trade because the price moves away. This costs 5.2 points of fill rate and 0.81 ticks.

- **Queue position.** The price reaches the order, but the queue ahead is not cleared in time. This costs 7.9 points and 0.85 ticks. The recorded queue state shows the effect directly: queue-aware fill rate falls from 93.8% with fewer than 200 shares ahead to 76.4% with 2,000 or more. Price-touch fill rates are flat or rising across the same buckets. 1,181 probes saw a trade at their own price and still did not fill; their median queue ahead was 900 shares, against 600 for probes that filled.
- **Adverse selection.** The negative queue-aware markout means the fills a real queue grants are disproportionately the ones where flow is running through the level.

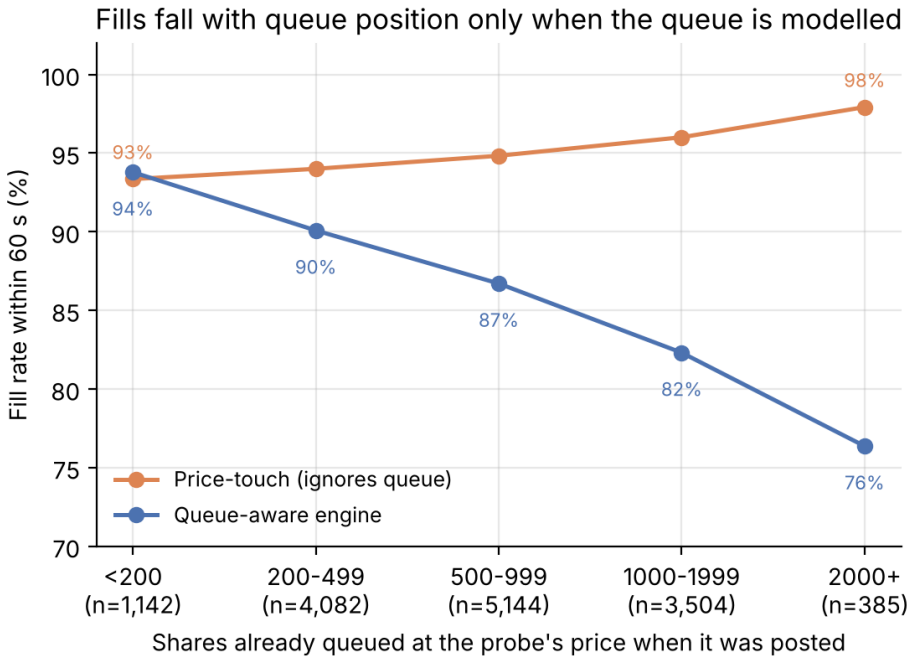


Figure 3. Fill rate within 60 s by shares queued ahead at posting, price-touch versus queue-aware.

The gap is present before any stress. In the calm pre-event window, the queue-aware effective spread is −0.40 (buys) and −0.21 (sells) against −1.32 naive. During the event, it concentrates on the side that benefits from the move:

- Asks resting while sellers sweep the bid filled 43% of the time (naive: 100%). The unfilled ones were completed 13.8 ticks below posted.
- The naive assumption books all of them as filled just before a 100-tick fall.
- Bids filled 100% under every assumption, with the same −18-tick markout. There, the sweep makes queue position irrelevant.

Identical flow, measured. The background event stream is identical across assumptions, but queue-aware probes are real orders and perturb the book:

- best quotes differ from the no-probe replay in 11.2% of seconds;
- 4.7% of decisions were posted at a different price;

- 31.7% of background fills are not reproduced exactly, mostly size shifts;
- own probes make up 4.4% of the queue ahead.

Restricting to decisions posted identically in both replays changes the headline by at most 0.04 ticks and 0.3 points.

6. Limitations

- **Synthetic only.** Real-data validation was attempted, but the free LOBSTER sample now requires a trial-account application (ADR-003).
- **Simple model.** One instrument; zero-intelligence flow with no informed traders, seasonality, or feedback from price to flow; limit orders never cross on arrival.
- **Simplified engine.** No hidden liquidity, auctions, latency, self-trade prevention or pro-rata matching.
- **Model-dependent results.** Both experiments' magnitudes depend on the calibrated parameters, chosen by hand against stated targets. The probe design (size, patience, placement) is a single choice.
- **Mechanical resilience.** Section 4 explains why the recovery speed in particular does not transfer.
- **One disclosed post-hoc change.** One fill-comparison diagnostic was replaced after the first full run because it double-counted. No pre-specified metric changed (ADR-008 addendum).

7. Reproducibility

`bash scripts/run_simulation.sh` runs the tests, the bug-reintroduction check and the full study, and writes every number above to `results/`. The runs are deterministic given the documented seeds; four full runs produced identical pre-specified results. Library versions are in `results/run_metadata.json`. The figures in this paper are produced by `python paper/make_figures.py` from the committed outputs in `results/`.

Data and code availability

All code, tests, the design record (`DECISIONS.md`) and every recorded output are available at https://github.com/mateeniqbal3/lob-simulator under the MIT License, and each release is archived on Zenodo (the DOI is listed in the repository README). No market data are used; all order flow is generated by the documented model from fixed seeds.

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